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Introduction:

This document describes the steps needed to load firmware into the Halo device.

1. Programming Setup:

- 1.1. Ensure the USB flash drive fixture (Item 921) is installed in the Halo Device Programming Fixture Assembly (Item 909) and the computer is powered on. Computer may be plugged into AC outlet as needed for battery capacity.
- 1.2. Open the folder on the flash drive and verify the software revision numbers, which will be displayed as part of the zip file name (Example: “sem_app_dfu_V_0_4_6.zip”), match the revision number of the controlled-copy software specs (Item 907 and 912).
- 1.3. If the software revision number does not match, remove all files and replace with the current revision by following the Assembly Procedure for the USB flash drive fixture (Item 921).
- 1.4. Save a copy of the .zip files on the USB drive to the desktop.
- 1.5. Record the assembly serial number and software versions/builds (all the numbers at the end of the hex file’s name, example: for sem_app_dfu_V_0_4_6.zip record 0.4.6) on the Test Record (Item 914)
- 1.6. Record the Halo PCB serial number on the Test Record (Item 914).

2. Initial Hardware Setup:

- 2.1. Hold the button down on the Halo device for 8 seconds until the device LED shines blue.
- 2.2. Connect USB micro cable (Item 905) between the Halo device and the programming computer (Item 909).
- 2.3. Using the Device Manager (Windows Key ->Device Manager), find the COM port the Halo device is connected to (it will be called nRF52).

3. Programming Procedure:

- 3.1. Open a command prompt from the start menu.
- 3.2. Enter the command “nrfutil dfu usb-serial -pkg <zip-file-name> -p <device-COM-port>” substituting the dfu file name of the software being loaded and the COM port determined in step 2.3
- 3.3. If this was the final device to be programmed, close the command prompt.

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- 3.4. Press the button on the front of the device for 3 seconds to power off the device.
- 3.5. Press the button on the front of the board for 3 seconds to power on the device.
- 3.6. Verify that the LED is blinking yellow and red if loading the application software or solid white for the test software.
- 3.7. Press the button on the front of the device for 3 seconds to power off the device.
- 3.8. Disconnect the USB micro to USB cable from the Halo device.
- 3.9. Once programming is completed, delete the software file from the desktop.
4. Have a second individual verify the software version loaded matches the software version on the released spec and complete the Test Record (Item 914).

Revision History

<u>Rev</u>	<u>Date</u>	<u>Change No.</u>	<u>Originator</u>	<u>Description of Change</u>
1	10-06-20	P-SEMP-008	D. Miller	Initial release.
2	11-06-20	P-SEMP-022	D. Miller	Updated to use dfu with nrfutil so updates are cryptographically signed. Update to use USB drive and computer as separate fixtures rather than fixture assembly.